

Tables

Table 1. Demographic and Audiometric Characteristics of the NHANES Cohort

Characteristic	Training (n=15,654 ears)	Test (n=3,914 ears)
Participants (n)	9,421	3,519
Age (years), mean (SD)	43.9 (14.4)	44.0 (14.4)
Age (years), median (IQR)	44 (31–56)	44 (31–56)
Female sex	50.9%	51.3%
Mean PTA-4 (dB HL)	13.3 (11.4)	13.3 (11.6)
Normal	88.0%	88.6%
Mild	8.8%	8.1%
Moderate	2.2%	2.3%
Moderately Severe	0.6%	0.7%
Severe	0.3%	0.3%
Profound	0.1%	0.0%

Table 2. Classification Performance on Held-Out Test Set

Metric	FAI (Fuzzy)	PTA-4 (WHO)	XGBoost	Random Forest
Weighted vs. Reference	0.73	1.00*	0.98	0.98
Overall Agreement	92.3%	100%*	98.8%	98.8%
Borderline Agreement (± 5 dB)	70.6%	100%*	93.5%	93.8%
Clear-case Agreement	96.9%	100%*	100%	100%
Spearman vs. PTA-4	0.58	1.00	1.00	1.00
MAE vs. PTA-4 (dB)	6.1	0.0*	0.2	0.5

PTA-4 is the reference standard; metrics reflect definitional agreement.

Table 3. Configuration Distribution in the NHANES Cohort (Exploratory)

Configuration	Prevalence (%)	Slope Range
Flat	56.7%	−5 to 12 dB
Gently Sloping	18.0%	12–28 dB
Steeply Sloping	8.8%	28–45 dB
Precipitous	3.5%	>45 dB
Rising	13.0%	<−5 dB

Table 4. Fuzzy vs. Crisp Classification at the Normal–Mild Boundary

PTA-4 (dB)	WHO Grade	FAI Score	FAI Grade	Key Memberships
24	Normal	9.3	Normal	Normal =0.29, Mild =0.00
26	Mild	9.3	Normal	Normal =0.15, Mild =0.00
27	Mild	14.8	Normal	Normal =0.07, Mild =0.25
28	Mild	17.8	Normal	Normal =0.00, Mild =0.50
30	Mild	20.4	Mild	Normal =0.00, Mild =1.00

Table 5. Optimised trapezoidal membership function parameters (a, b, c, d) for the six hearing loss severity categories.

Category	a	b	c	d
Normal	0.0	0.0	14.3	28.0
Mild	26.0	30.0	39.3	44.0
Moderate	41.0	45.0	53.6	59.0
Moderately Severe	56.0	60.0	67.9	74.0
Severe	71.0	75.0	85.4	94.0
Profound	91.9	94.4	103.5	120.0

Table 6. Comparator models evaluated on the held-out test set.

Model	Description	Target
PTA-4 (WHO)	Pure-tone average of 0.5, 1, 2, 4 kHz; classified using WHO crisp thresholds (25/40/55/70/90 dB)	Severity grade
XGBoost	Gradient-boosted regression trees (learning rate 0.1, max depth 6, 100 estimators); input = 7 frequency thresholds; output = continuous PTA-4 estimate	PTA-4 (continuous)
Random Forest	1,000 regression trees (max depth 10, min samples split 5); input = 7 frequency thresholds; output = continuous PTA-4 estimate	PTA-4 (continuous)

Table 7. Table (context: Classification Accuracy)

Method	vs. Reference	Borderline Agreement (± 5 dB)	Clear-case Agreement
FAI (fuzzy)	0.73	70.6%	96.9%
PTA-4 (WHO)	1.00*	100%*	100%*
XGBoost	0.98	93.5%	100%
Random Forest	0.98	93.8%	100%

PTA-4 is the reference standard; its agreement is definitional. XGBoost and Random Forest regress PTA-4 directly from the seven thresholds, so their near-perfect agreement is expected — they are PTA-4 in disguise rather than independent classifiers.

Table 8. Classification at the normal/mild boundary: crisp labels versus graded fuzzy output across the 24–30 dB HL range.

PTA-4	Crisp Label	Fuzzy Label	FAI Score	Normal	Mild
24 dB	Normal	Normal	9.3	0.29	0.00
26 dB	Mild	Normal	9.3	0.15	0.00

27 dB		Mild		Normal		14.8		0.07		0.25	
28 dB		Mild		Normal		17.8		0.00		0.50	
30 dB		Mild		Mild		20.4		0.00		1.00	

Table 9. Summary of the four clinical case studies (A–D).

Case		PTA-4		WHO Grade		FAI		Configuration		Key Finding	
A		27.5 dB		Mild		16.5		Precipitous (shape rules)		Borderline membership (Normal 0.04, Mild 0.38)	
B		23.0 dB		Normal		16.4		Precipitous (shape rules)		25 dB notch at 4 kHz masked by PTA	
C		33.8 dB		Mild		20.4		Sloping		Frequency-resolved severity preserved	
D		53.8/28.8 dB		Mod/Mild		56.5 composite		Precipitous		Asymmetry upgrade; inter-aural modulation	